2025 International Conference on Intelligent Systems and Pioneering Innovations in Robotics and Electric Mobility (INSPIRE) | 979-8-3315-8644-7/25/$31.00 ©2025 IEEE | DOI: 10.1109/INSPIRE67328.2025.11300620

ZKPass – A Zero-Knowledge Proof Based Authentication System for Secure Identity Verification

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***Abstract*—Secure digital identity management is every emerging concern. Traditional authentication which includes use of passwords, central databases, and third party recovery present great security and also usability issues. Zero- Knowledge Proofs (ZKPs) and blockchain have put forth as very good for Decentralized identity systems. But also many present solutions have large compute requirements, don't scale well, and have poor user recovery. This review puts forth that in present ZKP based identity systems we see the lack of password less login, human readable identities, and self sovereign recovery. We look at recent systems which we note have a heavy use of complex crypto credentials, central verification which is a point of failure, and extensive infrastructure which in turn do not see wide scale adoption. To present solutions to these issues we have put forth a UID based identity which uses ZKPs for authentication which does not require storage of passwords or private keys. We introduce a novel recovery which uses a human readable phrase from private key, salt, and UID which in turn is a user controlled method. What we did is we put the UID on the blockchain which in turn improves privacy and scale. Our analysis which we present improves on issues of usability, scale and security which in turn we present a very simple and practical solution for today's identity management issues. Also this study we present which we put forth to be the base for what we think will be future works in the development of useable ZKP based identity systems.**

***Index Terms*—Zero Knowledge Proof (ZKP), Decentralized Identity, Self Sovereign Identity, UID Based Authentication, Blockchain Security.**

I. Introduction

A. Background and Motivation

In today's digital era we observe that people are releasing more of their private information on line into different platforms that include social media, eCommerce, health care, and government organizations. Even though we have witnessed expansion in this sector the majority of these platforms continue to employ the same traditional authentication procedures of user names and passwords that consequently raise concerns of privacy, security, and convenience. We see issues like data breaches, phishing, password re-use, and that our identity info is stored in one place which in turn leads to identity theft and data leaks [1]. To that end the security and crypto community has put forth advanced solutions like Zero Knowledge Proofs (ZKPs) and blockchain which in turn we are re thinking the identity management and verification processes.ZKPs which in turn allow people to prove out that they have certain things (which may be credential or that they are who they say they are) without giving out private behind the scenes information at the same time they which also put in place a decentralized and unchangeable infrastructure for what is stored in the blockchain that in turn breaks our from the control of central authorities also reduces very large scale security issues [2], [3]. We see also how these techs have made way for what we know as decentralized identity (DID) frameworks which in fact put digital identity back in the hands of the individual. But what we also see is that current solutions are either extremely complicated that therefore remain beyond the grasp of the common either individually or they are largely institutional in form which in turn is to the disadvantage of extensive use.

B. Problem Statement

Varying Zero-Knowledge Proof-Based (ZKP) are built to demonstrate identity; they usually experience problems that nullify their practical challenging to use and deploy. Many of these systems reliant upon profoundly intricate cryptography protocols that consume so much computational power, hence they are impractical for lightweight programs [4]. Also, specific systems non-userfriendly interfaces or technology knowledge to operate beyond the space of everyday functioning [5]. An area that remains underexplored is password-less login utilizing human-readable unique identifiers (UIDs), a straightforward and familiar approach that can be seamlessly integrated with existing platforms [6].

Additionally, account recovery presents a significant challenge in decentralized systems. Most ZKP-based solutions do not provide self-sovereign recovery, instead depending on trusted third parties, social recovery methods, or additional cryptographic credentials, which reintroduce central points of failure and reduce user autonomy. There's a clear gap in current research: no secure, passwordless, user-focused identity system using ZKP with easy recovery and minimal blockchain storage.

C. Contribution and Scope

This review paper aims to address the identified gap by exploring digital identity systems that employ ZeroKnowledge Proofs (ZKPs), particularly those integrated with blockchain technology. We conduct a comprehensive review of the existing literature to discern common architectural patterns and evaluate the strengths and weaknesses of current solutions [2]. Subsequently, we introduce and evaluate an innovative UID-based identity framework that utilizes ZKPs to enable secure logins without the necessity for passwords or stored private keys. The framework enhances user privacy by permitting authentication through non-sensitive data. It also increases security and trust as blockchain's decentralization is combined with the privacy aspects of ZKPs. The system offers cross-platform support and defends against threats like identity fraud. We analyze performance, usability, and multidisciplinary integration as key factors [7].

This system incorporates a human-readable recovery phrase, generated by combining elements of the UID, a private key, and a secret salt, allowing users to independently recover their identity. Unlike traditional systems, only the UID is stored on the blockchain, thereby reducing storage costs and minimizing data exposure [1].

Our paper coverage is:

- A discussion about blockchain-based and ZKP-based identity systems from literature.
- Identification of present research gaps with respect to usability, recovery, and scalability.
- A complete description and analysis of our proposed UID-based ZKP scheme.
- A comparative analysis with available solutions.

It contributes to widening possibilities for advancements towards lightweight, privacy-focused, easy-to-use digital identity systems consistent with principles for self-sovereign identity and decentralized identification.

II. RELATED WORK AND EXISTING APPROACHES

A. Traditional Identity Management Models

The digital identity management field has seen great transformation, we went from centralized to federated identity models and most recently to Self Sovereign Identity (SSI) systems. Centralized identity which have a single authority for user credential management and identity authentication. Although such systems are easy to implement and very popular they also present scale issues which include privacy and security which in turn include identity theft and data breach [8]. Also central models present single points of failure which in turn may cause access compromise or large scale compromise [9], [10]. The comparison of literature with our study is shown in Table I.

Federated identity solutions also prescribe to some degree some of the problems mentioned above by enabling people to single sign into a number of applications. That is with protocols such as OAuth and SAML which encourage cross platform usage and thus help alleviate the paining which is surrounded with a myriad of unmanaged accounts [1], [13]. The problem is that federated systems still rely onto systems which, with a claim to safeguard our privacy, do very little to inform people that their information and behaviors will be aggregated and exposed to a number of services.

The Self Sovereign Identity (SSI) framework's increase in popularity has reliance on full control by the users pertaining to their digital identity. There are models like SSI that use Decentralized Identifiers (DIDs) and verifiable credentials on certain block chain networks that remove trust to the center and increase user control. In theory. this is very nice, but currently in practice, SSI systems implement very complicated systems of credential management, forcing users to engage in high-level crypto operations. Complexity and the lack of easy recovery and many other options is the reason why the system has not been widely adopted yet.

B. Zero-Knowledge Proofs in Identity Systems

Cryptography related work reports on Zero Knowledge Proofs (ZKPs) in which parties present on to each other certified info without transacting relevant data [14], [15]. For privacy based identity checkups ZKPs see great improvement from this feature of the algorithm. Also we see that some zk-SNARK and STARK which are types of ZKP algorithms put forth very accurate and small proofs which in turn are very easy to do a computerized check of via simple bit evaluation [16]. ZKPs have played a role in many identity related issues which include the verification of certain facts like age or residence without giving out private info [17] also we see their use in the verification of users without them to reveal passwords or biometric data [18] also in the provision of anonymous services [19], [20].

Yet in practice what we see are issues which we must overcome for implementation of these technologies. A great many techniques require large initial outlay [21] of resources, need high amounts of processing power or have very high barrier to entry for the intended users. Also it is mainly within the domain of designed systems at the organizational level that ZKP is applied, rather in the case of consumer grade systems which require simple and uncomplex interaction by the end user that it is deficient [22].

Account recovery is still a problem for systems. Most seem to choose social recovery protocol, trustworthy third party service providers, or partial cryptography key that reconsolidates

TABLE I
COMPARISON STUDIES

Aspect	Study [2]	Study [6]	Study [11]	Study [12]	Our Paper
Primary Focus	ZKP with SSI on blockchain	ZK-SNARKs for cloud	Anonymous auth in VANETs	Compact ZKP in blockchain	UID auth with lightweight ZKP
User-Centric Design	Low usability focus	Security focus, no UX	Moderate with anonymity	No usability features	Simple UI, easy recovery
Recovery Mechanism	Not covered	Not discussed	Not included	Not provided	Secret phrase recovery
Blockchain Utilization	SSI blockchain for identity	Data/proofs onchain	Auth & data sharing	Private transactions	UID on/off-chain, ZKP for speed
Technical Complexity	Moderate to high	High (ZK-SNARKs)	Moderate	Medium	Low, SHA-256 with UID+ZKP
Future Orientation	Enhance SSI with ZKP	Cloud privacy	Vehicular privacy	Efficient blockchain	Decentralized, scalable auth

the system by reducing user agency. In comparison, we employ a streamlined ZKP methodology that incorporates human-readable, clear identifier, with easily convoluted cryptography hash. The process minimizes the barriers of account recovery, and many computational and systemic complexities.

C. Blockchain-based digital identity management

It has seen remarkable improvements in the cultivation of decentralized identity solutions that we see in the case of immutability, transparency, and distributed consensus [23]. Also we have the benefit of no central third parties which in turn improves security for identity credential management and which also puts identities in a decentralized state. Also we see the feature of tamper resistance and the ability to run smooth systems [9].

Decentralized identity systems which issue in blockchains the registration and resolution of decentralized identifiers, to include public keys in and to run revocation registries [1].Also they improve security and trust by the distribution of control among many participants. But what we see is that many issues are putting their wide scale adoption on hold. Many blockchain based identity systems require large scale infrastructure which in turn may be present in private blockchains or special networks which at the same time reduce inter and intra system communication as well as which also increase in implementation cost.

On-chain storage of blockchain proofs and credentials is accompanied by some issues of scalability and privacy due to public chains immutability. On top of that, a necessity for users to handle cryptographic keys directly as well as to interact with several smart contracts and off-chain systems has a negative impact upon overall user experience. Use of blockchain-based identity systems which is most taken into account is indeed in organizations but never in consumer applications.

We have put forth a design which uses the UID on chain with everything else going off chain. Our approach we put forth that which reduces the blockchain footprint, we also reduce transaction costs, improve privacy, at the same time do not trade off against decentralization or security which in turn presents a scalable and user friendly identity management solution.

D. ZKPs in blockchain

The use of zero knowledge proofs has grown in blockchains to improve anonymity and for that which is secure, trustless credential validation. In identity management they are used to verify ownership of attributes or identity assertions without the actual data's release [1]. While many systems have implemented zk-SNARKs or zk- STARKs for private authentication we see that for the most part these are complex in crypto requirements, have a need for a setup, put out large scale processing needs which in turn makes these proof systems a no go for real time use or lightweight environments [24].

Many present ZKP based identity systems which put proofs and related data on chain what in turn increases blockchain load, transaction fees, and also what we see is privacy issue. Also it is put forth by these that they are to large for individual use rather than the average person [9]. In that sense the put forth by this paper framework is a minimal ZKP which does away with heavy crypto and on we are able to reduce complexity and cost at the same time chain proof storage. By instead of on chain we go off chain for proof generation and we put only the UID on the chain we maintain privacy and decentralization we are thus addressing what we as the main issues in past implementations [2].

III. PROPOSED UID-BASED ZERO-KNOWLEDGE FRAMEWORK

Recent we have seen the introduction of UID based zero knowledge proof (ZKP) identity frameworks which address issues present in current ZKP based authentication systems which include use of stored secrets, low user access, and centralization in recovery. These frameworks we have designed to put forth Decentralized, user friendly and secure authentication solutions which also include minimal on-chain data and passwordless login.

A. Architecture Overview

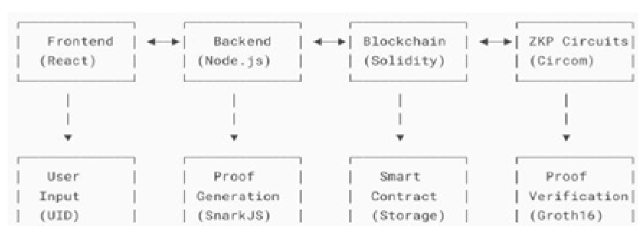

Fig. 1. Architecture diagram of ZKPass

A typical in which we see the implementation of UID based ZKP identity models which put forth a very light weight zero knowledge proof with blockchain smart contracts, in that we see a focus on human readable Unique Identifiers (UID) [14]. The architecture diagram is shown in Fig. 1. As opposed to what we have in present day ZKP systems which use complex crypto based identifiers or third-party identity providers, these frameworks put forth authentication via easy to remember UIDs [23]. Under such a convenience policy, such systems are being created with some basic building components. It comprises:

1) Client-side Interface: An easy-to-use interface accepts UID, invokes local computation to acquire a secret key and generation of zk-SNARK proofs with/without the use of users' cryptography knowledge [11].
2) ZKP Circuit Engine: In order to prove knowledge of something related to the UID without disclosing any secret information, proof generation software like Circom and Groth16 are typically employed [2].
3) Smart Contract: Smart contracts executing on Ethereum-like blockchains like Polygon execute on-chain verification of the zkSNARK proofs but maintain anonymity by only storing the UID hashes on-chain [21].
4) Backend Coordination: Its coordination via backend systems employs stateless RESTful APIs. These are commonly used in serving as a depository between the blockchain network and the user interface. They perform such operations as request verification and session handling while at the same time attaining even-tempered architecture systems without the storage of session authentication data, therefore increasing system data confidentiality and decentralization [18].

B. UID-Based Stateless Authentication

Unlike ZKP-based augmented authorization of the old days that demanded users to own crypto key pairs or rely on OAuth tokens, UID-based ZKP protocols work at their best with non-session-based mechanisms for generating shortterm key pairs [3]. For instance, some significant key generation algorithms, which rely heavily on character preferences and crypto salts, can derive private keys directly from UIDs [12]. These approaches produce keys only at the time of identification, without storing them elsewhere [23]. The registration workflow diagram is shown in Fig. 2.

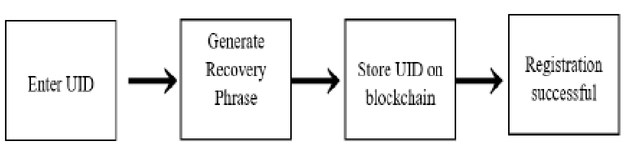

Fig. 2. Registration workflow diagram for ZKPass

Private key sets allow the owners to construct ZKSNARK proofs with the basic properties of zero-knowledge proofs remaining intact: completeness, soundness, and zeroknowledge [2]. The users that generate these proofs commit them to the blockchain, in which the blockchain's smart contracts check the proofs, consequently permitting user authentication without compromising sensitive information [21]. The verification model puts forward the emergence of going passwordless, protection against phishing and replay attacking, and seamless extension to web and mobile application [1].

C. Human-Readable Recovery Mechanisms

In self-sovereign architectures, the crucial challenge lies in self-recovery and the implementation of warm UID unlinkable ZKP systems. Much of the hype around this area arises from haphazard recovery statements that are not actually maintained within the system [9]. The recovery mechanism is based on salted UID information, which is translated into English word sets from a chosen vocabulary at a relatively simple phrase-disambiguation level [23]. This information is generated once and stored offline in a secure manner. During recovery mode, a predefined sentence is typed, which is then utilised to decompose the UID. The UID is constructed by superimposing elements and subsequently reducing the entropy of the ensemble, ultimately arriving at a principal UID [16].

IV. SECURITY AND PRIVACY

The UID based Zero-Knowledge Proof (ZKP) identity framework addresses key security and privacy challenges in modern authentication systems. It eliminates reliance on stored secrets and centralised recovery, offering a self-sovereign, privacy focused, and user friendly model. The primary security and privacy advantages include:

1) Elimination of Passwords or Stored Private Keys: Contemporary systems, such as zkLogin [14] and ZKBoo [24], require the storage of passwords or the management of long-term keys, thereby expanding the attack surface and increasing the vulnerability to threats such as credentials theft, phishing, and session hijacking. This framework enhances security by generating private keys locally and temporarily from human-readable UIDs during each login, thus obviating the need to store or transmit sensitive information.
2) Zero Exposure of Sensitive Data: Although systems such as Semaphore and Tornado Cash [19] prioritize anonymity, they offer limited or externally dependent recovery options. In contrast, the proposed framework introduces a deterministic, humanreadable recovery phrase that is generated during registration and is never stored. All recovery operations are conducted exclusively on the user's device, ensuring that no sensitive information is exposed or transmitted during the process of recovery.
3) No Need to Depend Upon Trustworthy Third Parties: For the core identity providers such as Google or GitHub that we see in zkLogin [14] that become points of single failure here is suggested another solution. Identity proofs are built locally and verification is observed to be handled by decentralized smart contracts that thus remove their dependency upon federated login services or third-party assertions.
4) Low On-Chain Activity with Good Privacy: For long term identity models based on advanced zero-knowledge protocols such as Semaphore [24] and zk-ID [4], large amounts of information, such as Merkle Roots or Public Keys are inserted into the blockchain. This approach results in higher gas costs and introduces what can

be described as identity unicity. To mitigate this, we propose maintaining only a hash of the on-chain UID. This method significantly reduces transaction volume and removes personally identifiable information, thereby lowering system-wide operational costs and reducing blockchain congestion.

5) Lightweight, Scalable ZKP for Real-World Applications: The usage of high-level ZKPs such as ZKBoo and Groth16 involves complex problem setups that require advanced cryptographic infrastructure. In contrast, less complicated ZKPs, such as those used in social logins and web dashboard circuits, can be effectively managed by existing frameworks. This simplification enhances usability and user experience while introducing minimal integration overhead with currently deployed systems. Furthermore, the system's design ensures efficient data flow ("system piping"), allowing even time-sensitive operations to be executed with little to no delay.

V. INTERPRETATION AND ANALYSIS

An overview of today's blockchain-based systems reveals that Zero-Knowledge Proofs (ZKPs) have been widely utilized across industries to address critical challenges related to verification, privacy, and data integrity. Amongst reviewed systems/schemes, especially noted are: SymmeProof [12], zkAuthFeed [23], LedgerMaze [21], HyperMaze [9], BAuth-ZKP [23], as well as ZAMA [19]. These systems utilize ZKP protocols within their enhancement with respect to safeguarding security as well as enhancing privacy concerning verified transactions, authenticated data feed transfers, multifactor verification, as well as mutual anonymous verification.

In regard to Table II, The focus of the rest of the analysis is on the proposed UID-ZKP system, while all other systems reviewed and their gaps remain as is in the current research. None of the systems which have been build so far tackle and independent multidimensional identities and authentication structure unobstructed frameworks systems user friendly design for civil level users. In particular self-sovereign account recovery, password-less login, blockchain storage efficiency less than 100 bytes per user record, low-cost computation resources, and ZKP wit's integrated seamlessly into system operation from the beginning.

The table describes multiple gaps that exist in the current implementation of ZKP-enabled blockchain identity systems. First up we see that present in all the systems reviewed is the use of passwordless logins which in turn supports the use of the traditional credentials which are a easy target for phishers and credential theft. On the other hand the put forth UID-ZKP system does not support password less auth but what it does is it provides a better and secure login experience. Also we note that we did not see self sovereignty in the account recovery which is a very important feature in all the studied systems. In the past we have seen that traditional systems which rely on central intermediaries or recovery agents for account recovery do raise issues related to privacy and trust.

The UID-ZKP system provides a means for identity recovery which is safe, requires no personal data to be shared, and no third-party services. With regard to blockchain storage, a considerable number of systems record transaction proofs or store considerable amounts of user data onchain. This approach not only increases the costs to maintain the blockchain and the stored data, but also poses an enormous risk to user privacy. In this respect, our system improves privacy and scalability by greatly reducing the onchain footprint and only preserving a unique user identifier (UID). From an effectiveness point of view, most systems nowadays have low performance since they rely on intricate cryptography systems. Our low-computationalresources- based methodology liberates portable devices from intensive power processing requirements. Edge devicespecially developed, the UID-ZKP system does away with low computation-based requirements for faster device access. enable faster access through edge devices since requirements for computation are extremely low.

To that which we put forth, although at this time ZKPs are very much in use, their application is in very specific areas like that of transaction authentication or selective attribute release. That which we present is UID-ZKP which breaks from full end to end ZKP implementation to put into practice extensive privacy features from the point of identity enrollment through to the point of authentication and authorization. Also as a whole we see that which this study brings to light the issues with the present structure and function of Block chain identity systems. We see that in present time these issues play out in the form of password based login, no self recovery feature, large scale on-chain data, high computational requirements, and lack of use of zero knowledge proofs which we solve with our put forth which at the same time provides great agility and extensibility as well as excellent privacy.

VI. DISCUSSION AND FUTURE DIRECTIONS

The Zero Knowledge Proof (ZKP) and identity systems on the blockchain are a great step forward for private user info. At the same time a great deal of issues still exist. Also these systems are very complex, require great computing resources, have limited data retrieval and do not fully implement ZKP. Some do provide decentralized identity which is a nice element but is very hard to work with, does not play well with other systems and fails at scale especially in the consumer space. Our study highlights the need for simple, user-friendly identity models that use password-free logins, allow self- recovery, and do not rely heavily on blockchain. Few systems address all these needs simultaneously. In addition, the limited use of ZKP in identity systems indicates a gap between theory and practice.

Future research should aim to create identity systems that are both secure and easy-to-use. Important areas to explore include improving compatibility with existing Self- Sovereign Identity (SSI) standards, examining post-quantum ZKP methods for long-term security, and designing easy-touse interfaces for non-experts. Testing real-world performance and ensuring

TABLE II
ZKP-BASED AUTHENTICATION SYSTEMS

Feature	SymmeProof [12]	zkAuthFeed [23]	LedgerMaze [21]	HyperMaze [9]	BAuthZKP	ZAMA [19]	UID-ZKP System
Passwordless login	X	X	X	X	X	X	✓
Self-recovery	X	X	X	X	X	X	✓
Blockchain storage	Partial	Partial	Full	Partial	Full	Partial	UID only
Computation required	High	Medium	High	Medium	Medium	Medium	Low
ZKP usage	Full	Full	Full	Full	Partial	Full	Full

compliance with rules such as GDPR will be the key to moving from academic models to real-world solutions.

VII. CONCLUSION

This review examined the current landscape of ZeroKnowledge Proof (ZKP) and blockchain-enabled identity systems, highlighting both their potential and the challenges they face. While existing frameworks have advanced privacy preserving authentication and decentralized identity management, they often fall short in terms of practical usability, lightweight design, and selfsovereign recovery options. Common challenges include high computational demands, reliance on centralized recovery methods, excessive on-chain data storage, and the limited integration of ZKPs in user authentication processes. Through a comparative analysis, this study identifies a significant gap in the literature: the absence of a fully integrated, user-friendly, and scalable ZKP-based identity model suitable for real- world applications. In response, the reviewed UID-based ZKP framework illustrates how password-less login, human- readable self-recovery, and minimal blockchain footprint can be achieved within a cohesive, decentralized system. The insights from this review suggest that future research in this field should aim to bridge the gap between cryptographic innovation and user experience. Interoperability with existing SSI standards, quantum-resilient protocols, and real-world validation will be crucial for developing the next generation of secure, privacy-preserving digital identity systems.

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